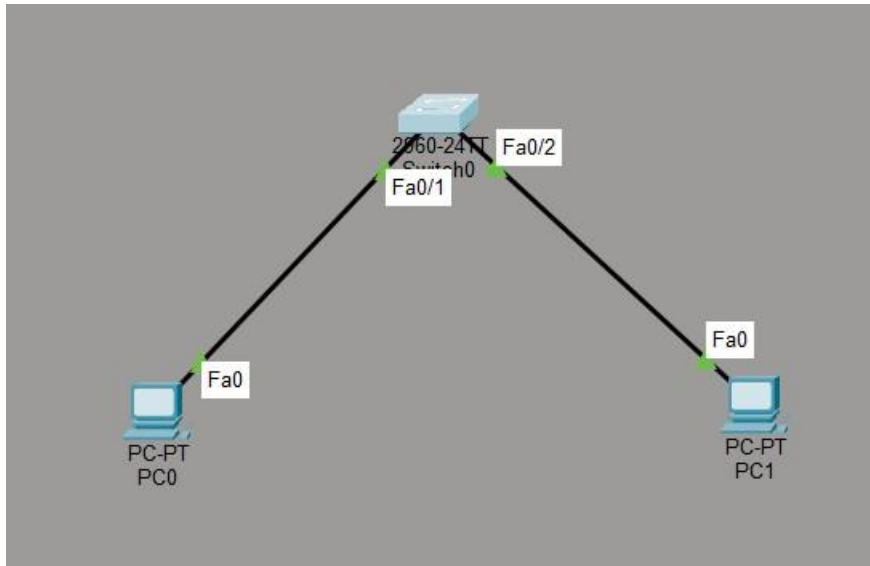


Task-1

Q- Open Cisco Packet Tracer or any network simulator, create a simple network with one switch and two PCs, and use the 'show mac address-table' command to observe how the MAC address table is initially empty.

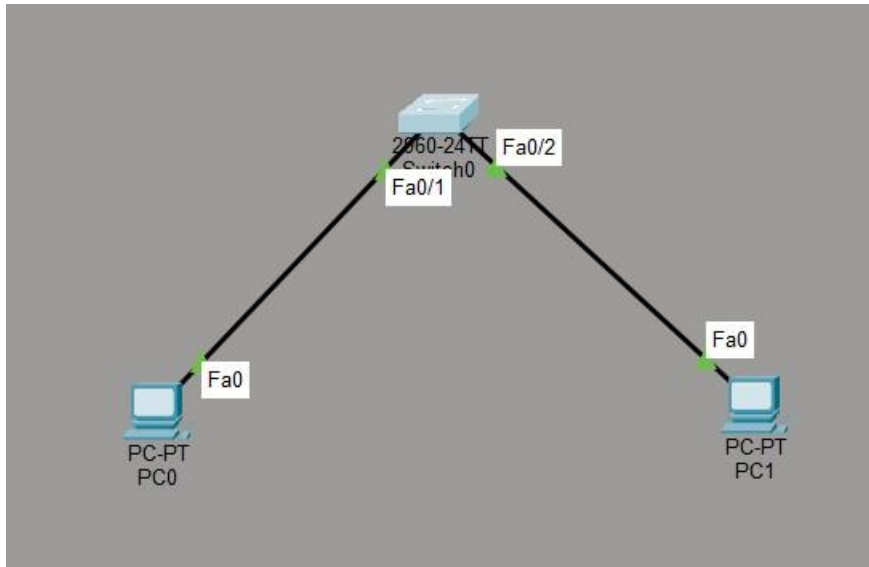


```
Switch>
Switch>en
Switch#enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
Switch#
```

Initially, the switch does not know which MAC address belongs to which port. When frames pass through the switch, it learns the source MAC address and stores it in the MAC address table along with the incoming port.

Task-2

Q- Send a ping from PC1 to PC2 in your simulated network, then immediately run 'show mac address-table' on the switch and note how the MAC addresses of both PCs appear in the table.
Hint: Try pinging in both directions to see how the table updates.

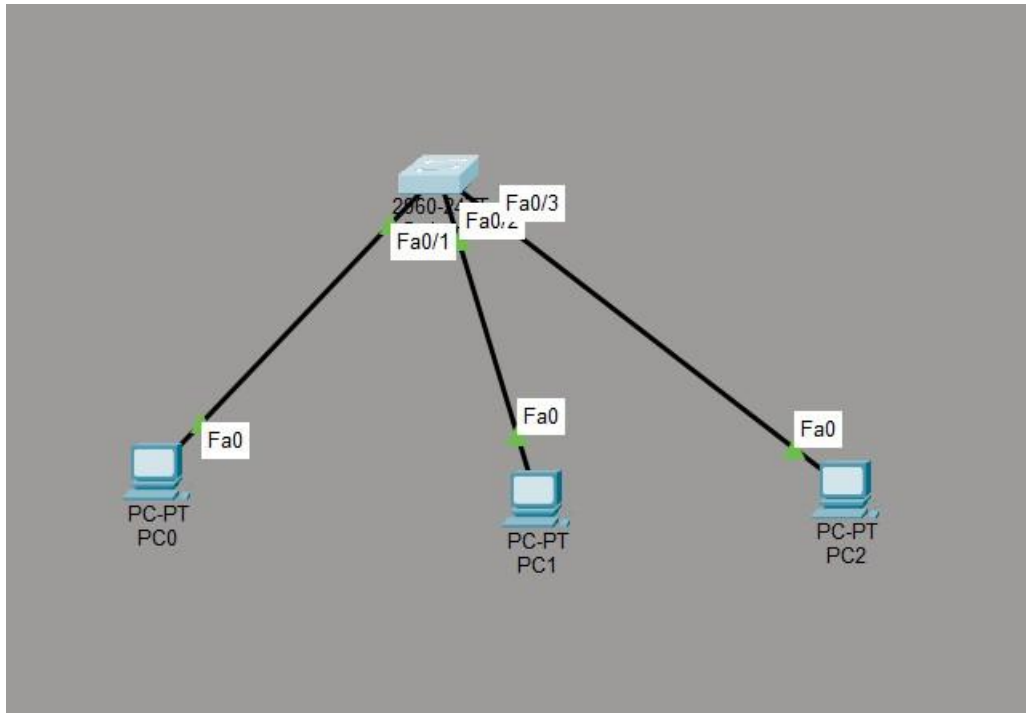


```
Switch>en
Switch#enable
Switch#show mac address-table
      Mac Address Table
-----
Vlan  Mac Address      Type      Ports
----  -
  1    0001.631a.130e    DYNAMIC   Fa0/1
  1    00d0.97ad.a23a    DYNAMIC   Fa0/2
Switch#
```

The experiment was successfully performed. The switch initially had no learned dynamic MAC entries for the two PCs. After sending ping traffic, the switch automatically learned the source MAC addresses and stored them in its MAC address table along with the corresponding switch ports.

Task-3

Q- Add a third PC to your network, connect it to the switch, and ping from PC3 to PC1. Observe and record how the MAC address table changes after this communication.



```

Switch>en
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.631a.130e    DYNAMIC Fa0/1
1       00e0.a3ae.3e09    DYNAMIC Fa0/3
Switch#show mac address-table
      Mac Address Table
-----
Vlan    Mac Address      Type    Ports
----    -
1       0001.631a.130e    DYNAMIC Fa0/1
1       00d0.97ad.a23a    DYNAMIC Fa0/2
1       00e0.a3ae.3e09    DYNAMIC Fa0/3
Switch#
  
```

Before PC3 communicated with the network, its MAC address was not present in the switch's dynamic MAC address table.

After PC3 sent a ping to PC1, the switch received frames from PC3 and learned its source MAC address.

The ping from PC3 to PC1 was successful. After the communication, the switch automatically learned PC3's MAC address and added it to the MAC address table with Fa0/3.

Task-4

Step-by-Step Frame Forwarding

Given MAC table:

Port	MAC Address
------	-------------

Port 1	00:11:22:33:44:55
--------	-------------------

Port 2	66:77:88:99:AA:BB
--------	-------------------

When a frame arrives on **Port 1** with destination MAC 66:77:88:99:AA:BB:

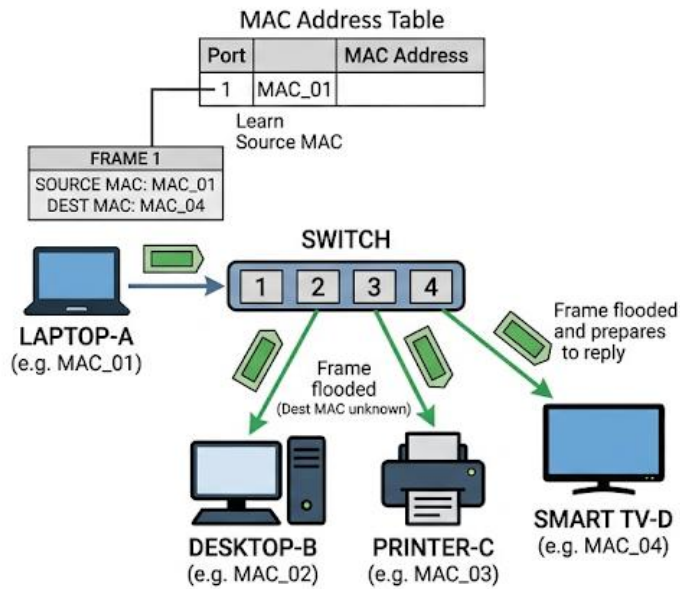
1. **Frame arrives:** The switch receives the frame through **Port 1**.
2. **Checks destination MAC:** The switch reads the destination MAC address: 66:77:88:99:AA:BB.
3. **Looks up MAC table:** The switch searches its MAC address table and finds this address associated with **Port 2**.
4. **Forwards the frame:** Because the destination is known, the switch sends the frame **only through Port 2**.
5. **Port 1 is not used for forwarding:** The switch does not send the frame back through Port 1 or broadcast it to other ports.

Result: The frame travels directly from **Port 1 → Port 2** because the switch already knows where the destination device is located.

task-5

Switch MAC Address Learning & Frame Forwarding: 4 Devices

Frame 1: Learning MAC Addresses



Frame 2: Using Known MAC Addresses

